Smart TimeTables

Clash-free college timetable generator, running entirely in the browser · Progress report, 21 October 2026

Name / Team: _____

Roll No.: _____

GNDEC Ludhiana

354	0	~0.5 s	94/100	83%	18/18
classes / week (GNDEC CSE)	clashes in 20 test runs	average generation time	average quality score	lab groups side by side	automated tests passing

PROBLEM & SOLUTION

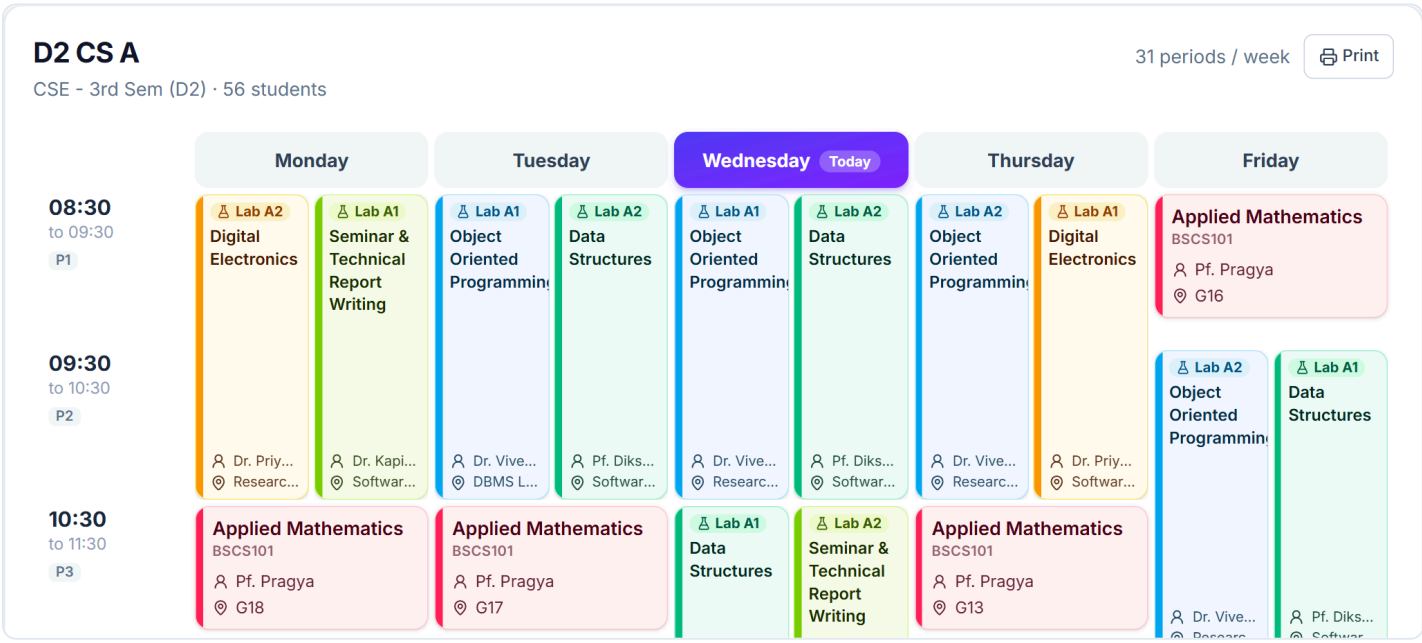
Timetables are made by hand over days and often contain clashes (a teacher in two rooms, a lab split by lunch). **Smart TimeTables** generates a complete weekly timetable for every section, teacher and room in about a second, with zero clashes, and gives each role its own view. It works for **any college** and is proven on **real GNDEC Ludhiana CSE data** (July–Dec 2026): 12 sections, 51 faculty, 19 rooms and labs.

SCHEDULING ENGINE

- **Hard rules:** no teacher, room or section double-booked; room size and type fit; labs in consecutive periods, never across lunch; faculty unavailability respected.
- **Preferences:** spread subjects across days, max 3 back-to-back periods per teacher, lab groups A1/A2 side by side.
- **Method:** pre-check of impossible input → most-constrained-first backtracking with busy grids → independent clash checker → 0–100 quality score. Clear plain-English reason if it can't fit.

FEATURES COMPLETED

- **Roles:** Administrator, HOD/Coordinator, Faculty, Student.
- **Setup wizard:** real clock times and lunch, rooms, subjects, faculty (with unavailable slots), sections with lab groups and separate lab teachers.
- **Views:** by section, teacher and room; HOD department filter; **faculty year filter** (D2/D3); **student lab-group filter**.
- **Design:** days across, times down; colour per subject; lab badges; today highlighted; subject legend.
- **Access:** light/dark theme, 3 text sizes, mobile layout, **Print** to one A4 page.
- **AI assistant:** "Ask AI" chat (Claude API) answers questions like "When is D3 CS A free on Wednesday?"
- **Data:** one-click GNDEC CSE and demo datasets; everything stored in the browser.



Generated timetable for D2 CS A (GNDEC data): lab groups A1/A2 run side by side in different labs (morning shown).

RESULTS

Dataset	Classes	Clashes	Avg. time	Quality
Demo college (10 runs)	114	0	39 ms	95–99
GNDEC CSE (10 runs)	354	0	508 ms	89–100

Stack: React 18, Vite, Tailwind CSS, Vitest, Claude API (serverless), Vercel. ~3,500 lines, built in 12 tested, committed steps.

NEXT STEPS

- Next Deploy on Vercel for a public link.
- Next Add the Claude API key to switch on the AI assistant.
- Add D4 (7th sem), cross-section electives and per-year lunch slots.
- Faculty time preferences; drag-and-drop edits; Excel import/export.